Pre-Analysis Plan

1 Hypothesis 1

1.1 Variables

We will include the following variables in the analysis:

Dependent variable: `gendereq` – an index measuring gender equality attitudes. This is constructed as the average of five items:

- `plh0301` (Women Should Rather Care About Family Than Career)
- `plh0303` (Men Involved in Housework, reversed)
- `plh0307` (Men Care more About Financial Security)
- `plh0308_v1` (Men and Women equal in Labor and Household/Family, reversed)
- `plh0309` (Working Mothers Equal Emotional Warmth, reversed)

Each item is measured on a 4-point Likert scale, in which 1 indicates "Agree Completely" and 4 indicates "Completely Disagree." Items are reversely coded as needed to ensure higher values correspond to more gender-equality. Items are set to missing if they take values below 0, indicating no available data. The index is computed as the average of the non-missing items. These variables are only available in the 2012 wave of the survey and are asked of respondents in cohorts L1, L2, and L3 only.

Main independent variable (endogenous): `prop_daughters` – the proportion of daughters among all biological children born before or during the survey year. We compute this using the variables `kidgeb01-kidgeb19` (year of birth of the i-th biological child) and `kidsex01-kidsex19` (gender of the i-th biological child). For each observation and survey year, we identify all children born up to that point, count the number of daughters, and divide by the total number of children to obtain the proportion.

Instrumental variable (IV): `first_child_female` – is a binary variable equal to 1 if the first child is female. This is computed from variable `kidsex01`.

Control variables:

- `sex` – recoded 1 for female, 0 for male.
- `partner` – coded 1 if the respondent is living with a partner, and 0 otherwise. This is based on the original `partner` variable, in which values 1–4 are taken to indicate living with a partner, 0 is taken to indicate not living with partner, and values less than 0 or equal to 5 indicate missing data.
- `germborn` – recoded 0 if born in Germany or immigrated before 1950, 1 if not born in Germany and missing otherwise.
- `westgerm` – coded 1 if sampreg (current sample region) takes the value 1 (West Germany), 0 if sampreg takes the value 2 (East Germany) and is missing otherwise.
- `pgbilzeit` – years of education; negative values are set to missing.
- `age` – age computed as survey year minus year of birth (`syear - gebjahr`).
- `age_sq` – age squared.
- `ln_hghinc` – natural logarithm of household income (ln of `hghinc`).
- `kids_0` to `kids_7`, and `kids_8plus` – total number of children dummies, computed from the variable `sumkids`.

No further recoding is applied unless indicated above.

1.2 Inclusion/exclusion criteria

Observations are excluded based on the following criteria:

- Respondents without biological children (`kidsex01` ≤ 0) are excluded.
- The sample is restricted to respondents from cohorts L1, L2, or L3 (i.e., those with `psample` values 12, 13, or 14).
- **Time period:** All available survey years are considered; however, only observations with a non-missing value on the `gendereq` index are included in the analysis. This applies exclusively to the 2012 wave, which is the only year in which the relevant gender attitudes items were collected.

1.3 Exact analysis

We estimate the following two-stage model using two-stage least squares (2SLS):

First stage (instrumenting the endogenous regressor):

$$\texttt{prop_daughters}_i = \pi_0 + \pi_1 \cdot \texttt{first_child_female}_i + X_i'\pi + u_i$$

Second stage (main regression equation):

$$\texttt{gendereq}_i = \beta_0 + \beta_1 \cdot \widehat{\texttt{prop_daughters}}_i + X_i'\gamma + \varepsilon_i$$

with:

- $\texttt{gendereq}_i$ is the index of gender equality attitudes for individual i.
- $\texttt{prop_daughters}_i$ is the proportion of daughters among all biological children, treated as endogenous.
- $\texttt{first_child_female}_i$ is a binary variable equal to 1 if the first child is female.
- X_i is a vector of control variables: `sex`, `partner`, `germborn`, `westgerm`, `pgbilzeit`, `age`, `age_sq`, `ln_hghinc`, and `kids_1` to `kids_7`. `kids_8plus` omitted because of collinearity.

We use robust standard errors.

1.4 Interpretation of results

To test the hypothesis that having daughters leads to stronger support for gender equality, we focus on the coefficient on `prop_daughters` in the second stage of the 2SLS regression.

- **Sign of $\hat{\beta}_1$**: A positive coefficient is in line with the hypothesis that having daughters leads to stronger support for gender equality.
- **Statistical significance of $\hat{\beta}_1$**: We use the robust z-statistic for $\hat{\beta}_1$ on `prop_daughters`. We apply a significance threshold of 5% (i.e., $p < 0.05$) to determine whether the effect is statistically distinguishable from zero.

A significantly positive $\hat{\beta}_1$ is interpreted as evidence in line with the hypothesis. If the coefficient is not significantly different from zero or is negative, we interpret this as no support for the hypothesis.

To evaluate the validity of the instrument (`first_child_female`), we also examine the first-stage F-statistic. An F-statistic greater than 10 indicates that the instrument is sufficiently strong, thereby supporting the reliability of the 2SLS estimates.

2 Hypothesis 2

2.1 Variables

We will include the following variables in the analysis:

Dependent variable: `lrscale` – an 11-point scale measuring political preferences, ranging from 0 ("Far Left") to 10 ("Far Right"). This is based on the variable `plh0004` (Political Attitude). Values below 0 are set to missing.

Main independent variable (DiD interaction term):
`first_child_female` × `post` – interaction between a binary variable whether the respondent's first child is female (`first_child_female` – generated from `kidsex01`) and a post-treatment indicator (`post`) equal to 1 for survey years after the birth of the first child.

Control variables:

- `sex` – recoded 1 for female, 0 for male.
- `partner` – coded 1 if the respondent is living with a partner, and 0 otherwise. Based on original `partner` variable, values less than 0 or equal to 5 are set to missing.
- `germborn` – recoded 0 if born in Germany, 1 if not born in Germany and missing otherwise.
- `westgerm` – coded 1 if living in West Germany (`sampreg` = 1), 0 if in East Germany (`sampreg` = 2), and missing otherwise.
- `pgbilzeit` – years of education; negative values are set to missing.
- `age` – age computed as survey year minus year of birth (`syear - gebjahr`).

- `age_sq` – age squared.
- `ln_hghinc` – natural logarithm of household income (ln of `hghinc`).
- `kids_0` to `kids_7`, and `kids_8plus` – total number of children dummies, based on the variable `sumkids`.
- Fixed effects: individual fixed effects (`pid`) and time fixed effects (`syear`)

2.2 Inclusion/exclusion criteria

Observations are excluded based on the following criteria:

- Respondents without biological children (`kidsex01` ≤ 0) are excluded.
- Observations are only used if the dependent variable `lrscale` is non-missing (i.e., `plh0004` ≥ 0).
- No survey years or cohorts are specifically excluded.

2.3 Exact analysis

We estimate the following Difference-in-Differences (DiD) model with individual and year fixed effects:

$$\texttt{lrscale}_{it} = \alpha_i + \lambda_t + \delta_1 \cdot \texttt{first_child_female}_i \times \texttt{post}_{it} + X'_{it}\beta + \varepsilon_{it}$$

with:

- $\texttt{lrscale}_{it}$ is the left-right political self-placement of individual i in year t.
- $\texttt{first_child_female}_i$ is a binary variable equal to 1 if the first child is female.
- $\texttt{post}_{it}$ is a binary indicator equal to 1 in the year of or after first childbirth, and 0 before.
- X_{it} includes control variables: `sex`, `partner`, `germborn`, `westgerm`, `pgbilzeit`, `age`, `age_sq`, `ln_hghinc`, and child-number dummies.
- α_i and λ_t denote individual and year fixed effects.

We cluster standard errors at the individual level.

2.4 Interpretation of results

To test the hypothesis that having daughters leads to more left-leaning political preferences, we focus on the coefficient δ_1 on the interaction term `first_child_female` $\times$ `post`.

- **Sign of $\hat{\delta}_1$**: A negative coefficient indicates that having a daughter is associated with more left-leaning political preferences.
- **Statistical significance of $\hat{\delta}_1$**: We use a t-test with clustered standard errors. A p-value below 0.05 is considered statistically significant.

A significantly negative $\hat{\delta}_1$ is interpreted as evidence in line with the hypothesis. If the coefficient is not significantly different from zero or is positive, we interpret this as no support for the hypothesis.

3 Hypothesis 3

3.1 Variables

We will include the following variables in the analysis:

Dependent variable: `share_work_alldays_partners` – the share of total hours of household labor performed by the respondent within a couple during the week. Total labor adds together hours spent on errands on weekdays (`pli0040`) and hours spent on housework and childcare on Saturdays (`pli0012_h`, `pli001_h`), Sundays (`pli0016_h`, `pli0022_h`), and weekdays (`pli0043_h`, `pli0044_h`). Values for weekdays are multiplied by five to estimate the weekly hours of labor. Partner labor is matched based on partner identifiers and the share is computed only when both partners' labor totals are available.

Main independent variable (DiD interaction term):
`first_child_female` $\times$ `post` – interaction between a binary variable whether the respondent's first child is female (`first_child_female` – generated from `kidsex01`) and a post-treatment indicator (`post`) equal to 1 for survey years after the birth of the first child.

Control variables: We include individual fixed effects (`pid`) and time fixed effects (`syear`).

3.2 Inclusion/exclusion criteria

Observations are excluded based on the following criteria:

- Respondents without biological children (`kidsex01` ≤ 0) are excluded.
- Respondents not living with their partner at the time of the survey (`pld0133` $\neq 1$) are excluded.
- Respondents with missing respondent ID (`pid`) or partner ID (`parid` = -2) are excluded.
- Observations must fall within the window of three years before and after the birth of the first child (i.e., $-3 \leq$ `rel_year` ≤ 3).
- Observations with missing values for either respondent or partner labor inputs are excluded.
- We only include male respondents (`sex` $= 0$) in the regression, so that we observe the change in the male share of household labor.

3.3 Exact analysis

We estimate the following Difference-in-Differences (DiD) model with individual and year fixed effects:

$$\texttt{share_work}_{it} = \alpha_i + \lambda_t + \delta_1 \cdot (\texttt{first_child_female}_i \times \texttt{post}_{it}) + \varepsilon_{it}$$

with:

- $\texttt{share_work}_{it}$ is the share of total household labor performed by the male respondent i in year t.
- $\texttt{first_child_female}_i$ is a binary variable equal to 1 if the first child is female.
- $\texttt{post}_{it}$ is a binary indicator equal to 1 in the year of or after first childbirth, and 0 before.
- α_i and λ_t denote individual and year fixed effects.

We cluster standard errors at the individual level.

3.4 Interpretation of results

To test the hypothesis that having daughters leads to more gender equality in the household, we focus on the coefficient δ_1 on the interaction term `first_child_female` $\times$ `post`.

- **Sign of $\hat{\delta}_1$**: A positive coefficient indicates that men who have a first-born daughter increase their share of household labor following birth, in line with the hypothesis.
- **Statistical significance of $\hat{\delta}_1$**: We use a t-test with clustered standard errors. A p-value below 0.05 is considered statistically significant.

A significantly positive $\hat{\delta}_1$ is interpreted as evidence in line with the hypothesis. If the coefficient is not significantly different from zero or is negative, we interpret this as no support for the hypothesis.

4 Hypothesis 4

4.1 Variables

We will include the following variables in the analysis:

Dependent variable: `gendereq` – an index measuring gender equality attitudes. This is constructed as the average of five items:

- `plh0301` (Women Should Rather Care About Family Than Career)
- `plh0303` (Men Involved in Housework, reversed)
- `plh0307` (Men Care More About Financial Security)
- `plh0308_v1` (Men and Women Equal in Labor and Household/Family, reversed)
- `plh0309` (Working Mothers Equal Emotional Warmth, reversed)

Each item is measured on a 4-point Likert scale, in which 1 indicates "Agree Completely" and 4 indicates "Completely Disagree." Items are reversely coded as needed to ensure higher values correspond to more gender-equality. Items are set to missing if they take values below 0, indicating no available data. The index is computed as the average of the non-missing items. These variables are only available in the 2012 wave of the survey and are asked of respondents in cohorts L1, L2, and L3.

Main independent variable (endogenous): `prop_daughters` – the proportion of daughters among all biological children born before or during the survey year. We compute this using the variables `kidgeb01-kidgeb19` (year of birth of the i-th biological child) and `kidsex01-kidsex19` (gender of the i-th biological child). For each observation and survey year, we identify all children born up to that point, count the number of daughters, and divide by the total number of children to obtain the proportion.

Instrumental variable (IV): `first_child_female` – is a binary variable equal to 1 if the first child is female. This is computed from the variable `kidsex01`.

Moderating variable: `sex` – a binary variable indicating respondent's gender, recoded 1 for female, 0 for male.

Control variables:

- `partner` – coded 1 if the respondent is living with a partner, and 0 otherwise. Based on original `partner` variable, values less than 0 or equal to 5 are set to missing.
- `germborn` – recoded 0 if born in Germany, 1 if not born in Germany and missing otherwise.
- `westgerm` – coded 1 if living in West Germany (`sampreg` = 1), 0 if in East Germany (`sampreg` = 2), and missing otherwise.
- `pgbilzeit` – years of education; negative values are set to missing.
- `age` – age computed as survey year minus year of birth (`syear - gebjahr`).
- `age_sq` – age squared.
- `ln_hghinc` – natural logarithm of household income (ln of `hghinc`).
- `kids_0` to `kids_7`, and `kids_8plus` – total number of children dummies, based on the variable `sumkids`.

4.2 Inclusion/exclusion criteria

Observations are excluded based on the following criteria:

- Respondents without biological children (`kidsex01` ≤ 0) are excluded.

- The sample is restricted to respondents from cohorts L1, L2, or L3 (i.e., those with `psample` values 12, 13, or 14).
- **Time period:** All available survey years are considered; however, only observations with a non-missing value on the `gendereq` index are included in the analysis. This applies exclusively to the 2012 wave, which is the only year in which the relevant gender attitudes items were collected.

4.3 Exact analysis

We estimate the following model using two-stage least squares (2SLS), allowing for heterogeneous treatment effects by respondent's gender via an interaction:

First stage:

$$\begin{aligned}\texttt{prop_daughters}_i = \pi_0 &+ \pi_1 \cdot \texttt{first_child_female}_i \\ &+ \pi_2 \cdot \texttt{first_child_female}_i \times \texttt{sex}_i + \pi_3 \cdot \texttt{sex}_i + \mathbf{X}_i'\delta + u_i\end{aligned}$$

$$\begin{aligned}\texttt{prop_daughters}_i \times \texttt{sex}_i = \theta_0 &+ \theta_1 \cdot \texttt{first_child_female}_i \\ &+ \theta_2 \cdot \texttt{first_child_female}_i \times \texttt{sex}_i + \theta_3 \cdot \texttt{sex}_i \\ &+ \mathbf{X}_i'\eta + v_i\end{aligned}$$

Second stage:

$$\begin{aligned}\texttt{gendereq}_i = \beta_0 &+ \beta_1 \cdot \widehat{\texttt{prop_daughters}}_i + \beta_2 \cdot \widehat{\texttt{prop_daughters}_i \times \texttt{sex}_i} \\ &+ \beta_3 \cdot \texttt{sex}_i + X_i'\gamma + \varepsilon_i\end{aligned}$$

with:

- $\texttt{gendereq}_i$ is the index of gender equality attitudes for individual i.
- $\texttt{prop_daughters}_i$ is the proportion of daughters.
- $\texttt{first_child_female}_i$ is a binary variable equal to 1 if the first child is female.
- $\texttt{sex}_i$ indicates whether the individual is female.
- X_i is a vector of the control variables listed above.

We use robust standard errors.

4.4 Interpretation of results

To test the hypothesis that having daughters leads to a larger increase in the support of gender equality for men than for women, we focus on the coefficient on the interaction term $\texttt{prop_daughters}_i \times \texttt{sex}_i$ in the second stage of the 2SLS regression.

- **Sign of $\hat{\beta}_2$**: A negative coefficient is in line with the hypothesis that having daughters leads to a larger increase in the support of gender equality for men than for women.
- **Statistical significance of $\hat{\beta}_2$**: We use the z-statistic for $\hat{\beta}_2$, with $p < 0.05$ considered statistically significant.

A significantly negative $\hat{\beta}_2$ is interpreted as evidence in line with the hypothesis. If the coefficient is not significantly different from zero or is positive, we interpret this as no support for the hypothesis. We assess instrument strength via the first-stage F-statistic.